

Business Needs generation — EDA

Question: are the generated Business Needs **grounded** (no invention) and **complete**, do they **use every selected stage**, and are each stage's needs **in-scope** for that stage?

What is generated

Business Needs are generated **per approved value stream**, for its **selected stages** (here: the GT stages), from the **raw idea card only** (no summary, no generation signals — the locked theme-gen decision). The output is one consolidated document structured as one block per stage:

```
Value Stage: <stage name>
  Business Product Feature: <scope area>
    1. <business need>  Note / Dependency / Business Rule (each only if stated)
    Operational Training / Operational Reporting  (only if stated)
  <repeat per selected stage>
```

How we evaluate — reference-free + structural

Like description, we do **not** score against the free-form GT text (matching it penalises style). We judge each value stream's Business Needs against the **raw source**, plus two **structural** checks the stage-keyed format makes possible:

metric	measures	vs
faithfulness	claims grounded in the idea card (no invention)	source
hallucination	1 - faithfulness (the invented claims)	source
coverage	the idea card's key facts reflected	source
stage_usage	selected stages addressed in the output	the stage set
stage_align	addressed stages whose needs fit that stage's scope (not misfiled)	catalogue scope

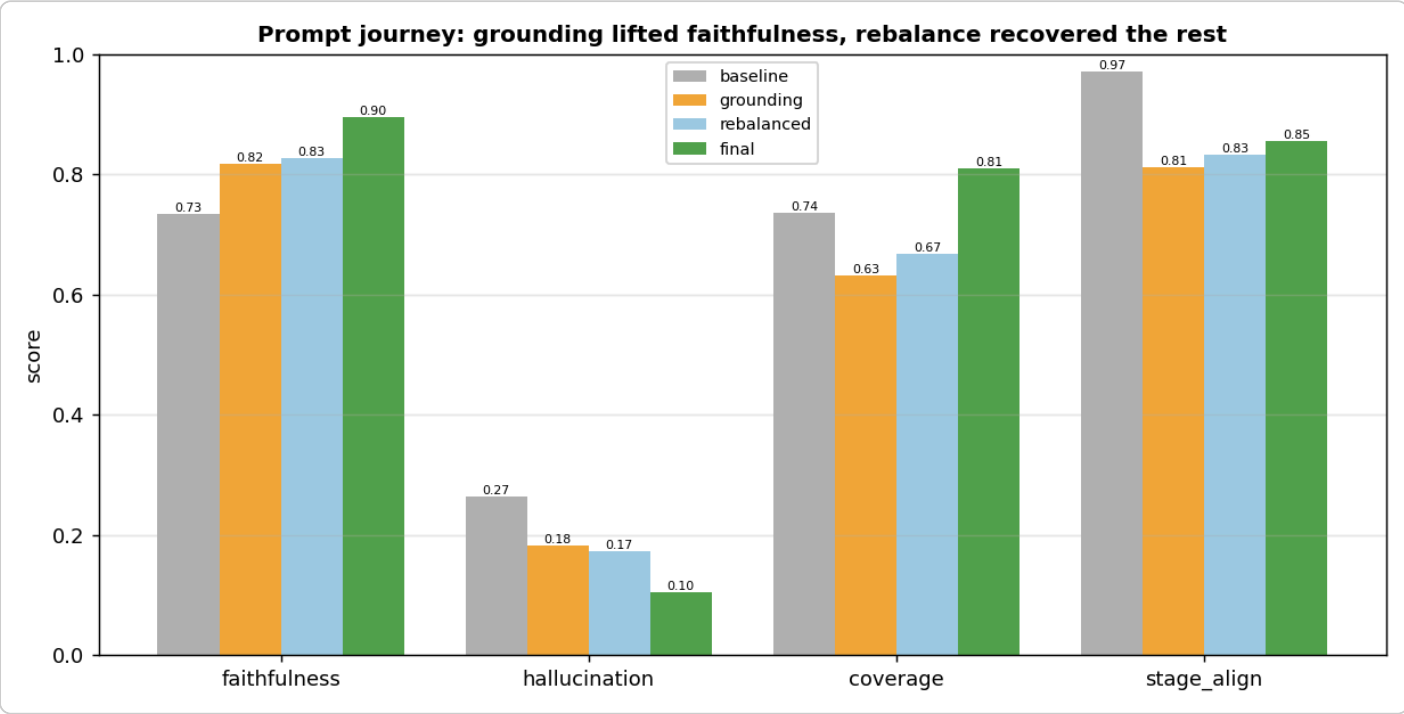
Where is "correctness"? There is no separate correctness metric, by design. Correctness-vs-GT was dropped because each GT Business Needs document is free-form (matching it would penalise style, not substance). In a reference-free eval, *"is this need correct?"* becomes *"is it supported by the source?"* — so **faithfulness IS the correctness check** (a need is correct iff the idea card backs it), and coverage is the completeness check. We keep faithfulness + hallucination + coverage rather than a redundant "correctness" label.

Finding — the structural checks pass; grounding is the lever

The **stage checks were excellent from the start** and held throughout: the model **uses every selected stage** and files needs under the right one.

metric	baseline	final
stage_usage	1.000	0.999
stage_align	0.972	0.855

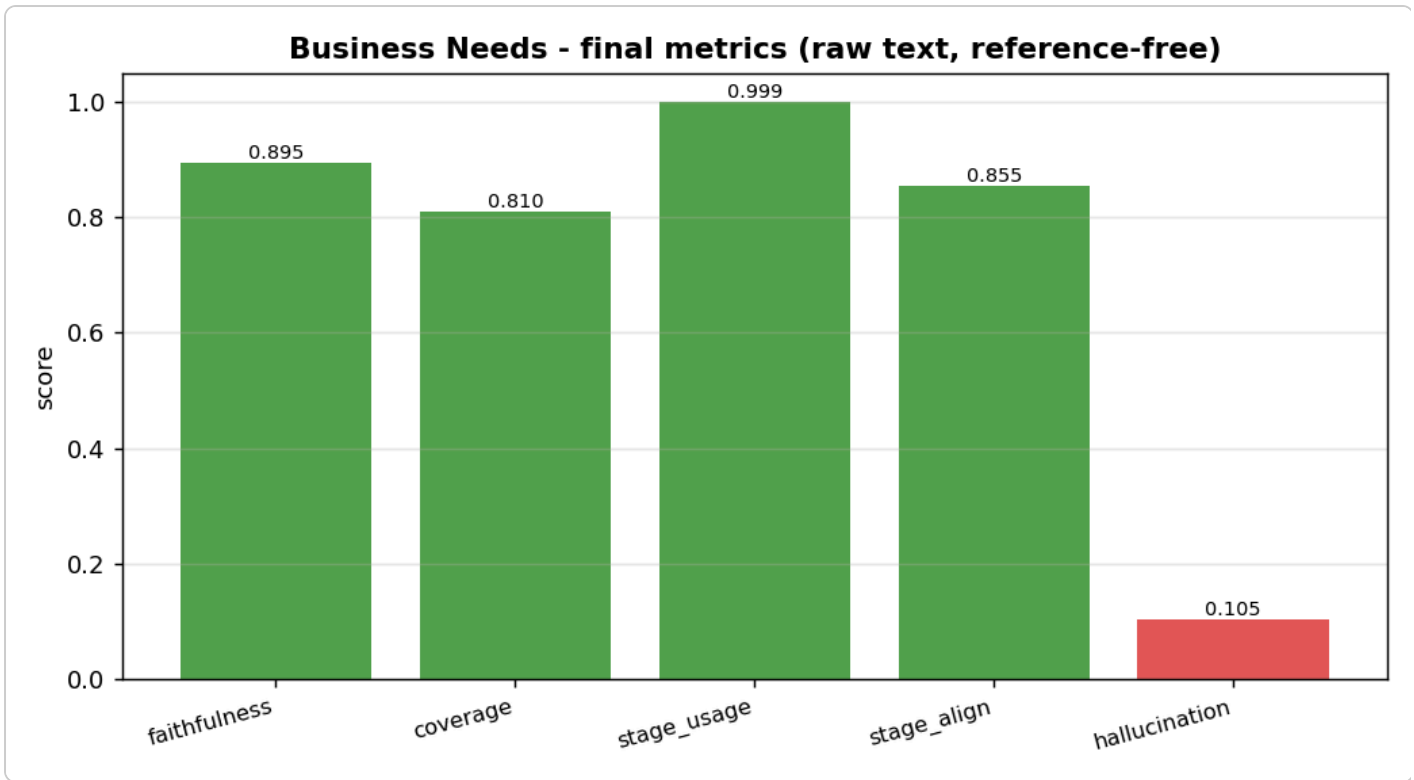
The real work was **grounding**. Business Needs is **prescriptive** ("the business needs X must be built"), so it invents requirements far more readily than descriptive prose — the baseline leaked 26% hallucination.



The journey — the prompt versions on the same sample, ending at the locked config:

metric	baseline	grounding (over-tight)	rebalanced	final
faithfulness	0.735	0.817	0.827	0.895
hallucination	0.265	0.183	0.173	0.105
coverage	0.736	0.633	0.669	0.810
stage_align	0.972	0.812	0.833	0.855

- 1. Diagnosis:** the prompt was **signal-centric** (Operational Training/Reporting "only if signals exist"; Note/Dependency/Business Rule) — but theme gen feeds **raw text with no signals**, so the model **fabricated** dependencies, business rules, and training/reporting from inference. 1 in 4 claims unsupported.
- 2. Grounding pass:** a hard rule (every need/note/dependency/rule must trace to a card phrase; conditional sub-fields reframed from "signals" to "the card explicitly states it; never infer") lifted faithfulness, but **over-corrected**: "a thin stage gets few needs / don't pad" made the per-stage needs **sparse and vague**, dropping coverage and stage_align — vague one-liners are hard to scope-match.
- 3. Rebalance:** reworted from "write less" to "write only what's grounded, but capture **everything** the card supports and state each need concretely enough to belong to *THAT* stage." This **kept the no-invention gain and recovered alignment and coverage**.



The honest read

- **Faithfulness 0.895 / hallucination 0.105** is strong for a **prescriptive** artifact — inventing requirements is its inherent failure mode, so ~0.10 hallucination is a good result on raw-only input.
- **Coverage 0.810** — grounding hard while still capturing what the card supports; the no-invention rule is the domain's #1, and the rebalance kept completeness alongside it.
- **The structural checks are the win you asked for:** stage_usage ≈ 1.0 (no stage silently dropped) and stage_align 0.855 (needs filed under the right stage) — the document is well-structured.

Verdict — locked

Business Needs: raw idea card only, grounding-rebalanced prompt → **faithfulness 0.895 / hallucination 0.105 / coverage 0.810 / stage_usage 0.999 / stage_align 0.855.**

- Reference-free (judged vs source; the GT format is too varied to score against), plus the stage-usage / stage-alignment structural checks.
- Grounding was the lever; the rebalance kept the no-invention gain while recovering stage alignment and coverage.
- Faithfulness 0.895 / hallucination 0.105 is strong for a prescriptive artifact, with the structural checks near-perfect (stage_usage ≈ 1.0 , stage_align 0.855).

No further changes.